

ConsultBae — AI Automation Take-Home Assignment

Deadline: 48 hours from the time you receive this email. Late submissions are not evaluated.

Expected effort: 10-14 focused hours across the 2 days. We are not looking for perfection, we are looking for working software and clear thinking. You WILL hit things you have never done before. That is intentional: figuring out what you do not know is the #1 skill this job needs.

The problem

At ConsultBae we run multiple products (recruitment gigs, CBNexus, and internal automations) and each has its own database. Your job is to merge them and automate on top, exactly the kind of work you would do here.

You are given 3 CSV files, each from a different system, with overlapping people and inconsistent, messy data. This data is intentionally imperfect. Part of your evaluation is how many problems you find and how sensibly you handle them.

Task 1 — Merge (core)

Design one clean database (SQLite/MySQL/Postgres, your choice) and write a script or pipeline that ingests all 3 files into it. The same person appearing in multiple files must become ONE record. Note: no single ID field is common across all files, you will need to think about how to match people.

Task 2 — Automate with a no-code/low-code tool (core)

Build ONE working automation using n8n (free, self-hosted or cloud trial), Make, or Zapier, connected to your data. Pick any ONE:

- A flow that receives a new CSV (webhook/email/drive trigger), checks it against your database, and sends a duplicate alert (email/Slack/webhook)
- A flow with an LLM step that auto-tags each person's skill category (e.g. "automation-heavy", "web dev", "data") and writes results back
- Your own idea of similar scope — surprise us

Export the flow JSON into your repo and show it running in your video. **Pure-code solutions for this task score zero**, we use these tools daily and need to see you can work no-code too.

Task 3 — Mini audio collection app (core)

We run gig projects where thousands of workers submit audio recordings. Build a miniature version: a web page (any stack, plain HTML/JS, Streamlit, Flask, React, your call) where a person can:

- Enter their name and phone number
- Record audio in the browser OR upload an audio file (either is fine)
- Submit — the audio gets stored, and a record goes into your database from Task 1

For every submitted audio, your app must automatically extract and store: duration, sample rate (kHz), bitrate, and loudness (dB). Bonus: a rough noise/quality estimate. If you have never worked with audio before, good, neither had we. Figure it out.

Plus a second view listing all submissions with a play button and the extracted properties. Ugly is fine; working matters. Deploy it anywhere free (Render/Railway/Streamlit Cloud/ngrok during your video) or demo it running locally in your recording.

Task 4 — Data issues report (core)

A list of every data quality problem you found in the 3 files and what you did about it. Be specific.

Task 5 — Stretch (optional, attempt if you can)

One page, no code: we launch your audio app to 5,000 gig workers over a single weekend. What breaks first? What would you change before launch? Think storage, uploads, failures, duplicates, cost.

Submission (all 5 required)

- **GitHub repo** with your code. Commit as you go, we look at commit history, not just the final code.
- **README** with setup steps and your data issues report.
- **Stuck log** (in the README): the 2-3 hardest places you got stuck, and exactly how you got unstuck, what you searched, what you asked AI, what suggestions you rejected and why. Blank or generic stuck logs score zero: everyone gets stuck on this assignment, and how you get unstuck is the main thing we are hiring for.
- **Screen recording (max 6 min)**, Loom or any tool): run your pipeline, show the audio app working end-to-end, walk through the 2-3 decisions you found hardest. Face not required, voice required.
- Reply to the assignment email with the repo link + video link before the deadline.

A note on AI tools

You may use ChatGPT/Claude/Copilot, we use them daily too. But you must understand and be able to defend every line. Shortlisted candidates will extend their own solution live on a call. In that round your submission is the starting point, and "the AI wrote it" is an instant no.

What we score

Working end-to-end pipeline over polish. Sensible matching logic. How many planted data issues you caught. Judgment in your stuck log and video. Speed.

Good luck. Ship it.